CMSC 27100-1	Due: Monday, November 4, 2024 (submit online via Gradescope!)

Problem Set 4

Instructor: Bill Fefferman

1 Practice with Induction

Problem 1: Prove that the following holds for all non-negative integers n and all real numbers $x \neq 1$:

$$\sum_{i=0}^{n} x^i = \frac{x^{n+1} - 1}{x - 1}$$

Problem 2: Recall that "n factorial" denoted $n!$, is defined for any positive integer n to be the product of all positive integers less than or equal to n. In other words $n! = n \cdot n - 1 \cdot n - 2 \cdot ... \cdot 1$. Prove that for integers $n \geq 2$:

$$n!2^n < (2n)!$$

Problem 3: Use strong induction to prove that every integer greater than 1 can be written as a product of primes. Be formal with your proof, indicating the base case and inductive hypothesis. Also explain why you used *strong* induction in this argument.

2 More number theory

Problem 4: Find the remainder of 67! when divided by 71, without using a computer. Explain your work, and here's a hint: Use Wilson's theorem (note, we'll cover the theorem this week).